



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VIYAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

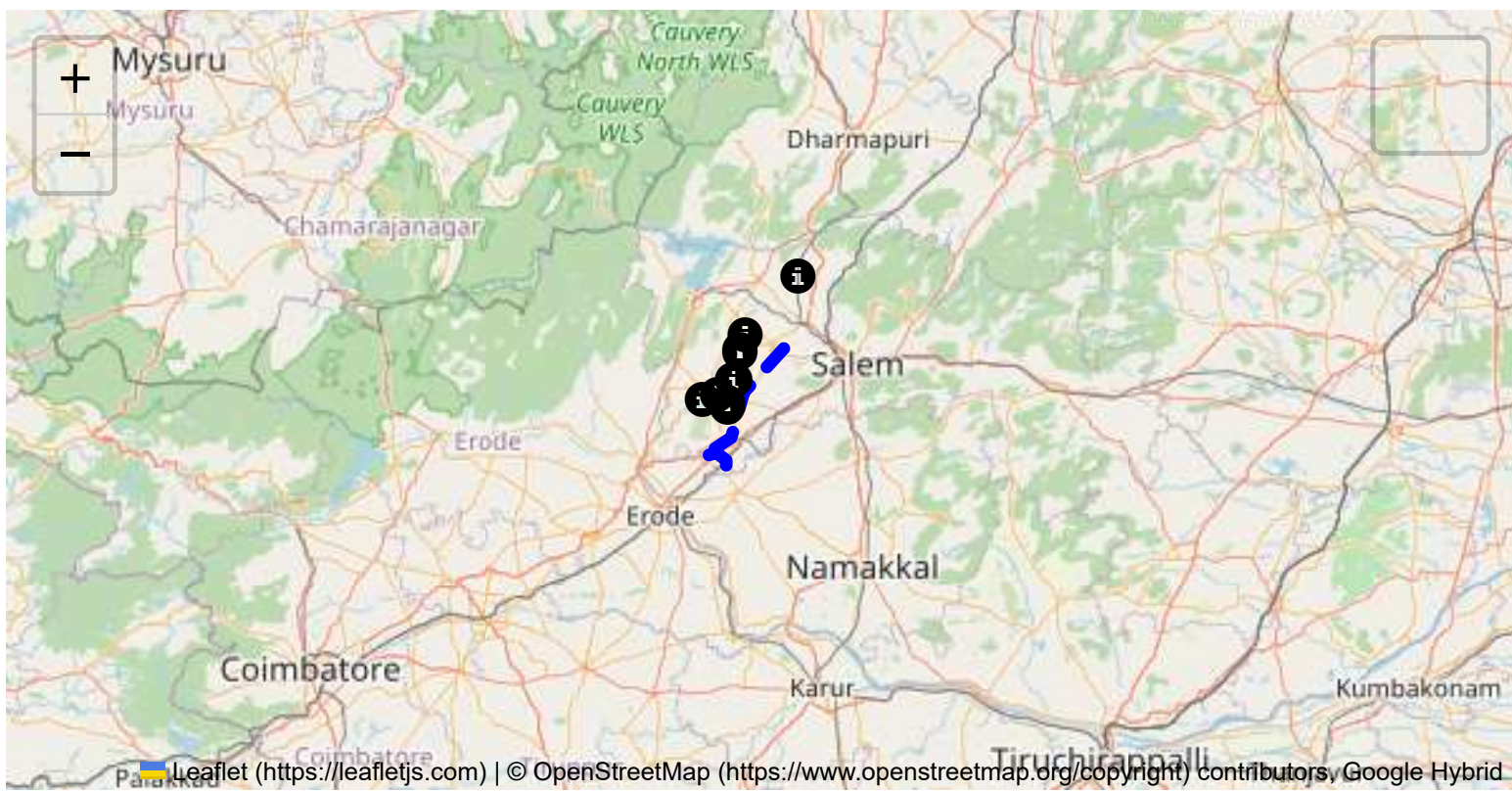
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

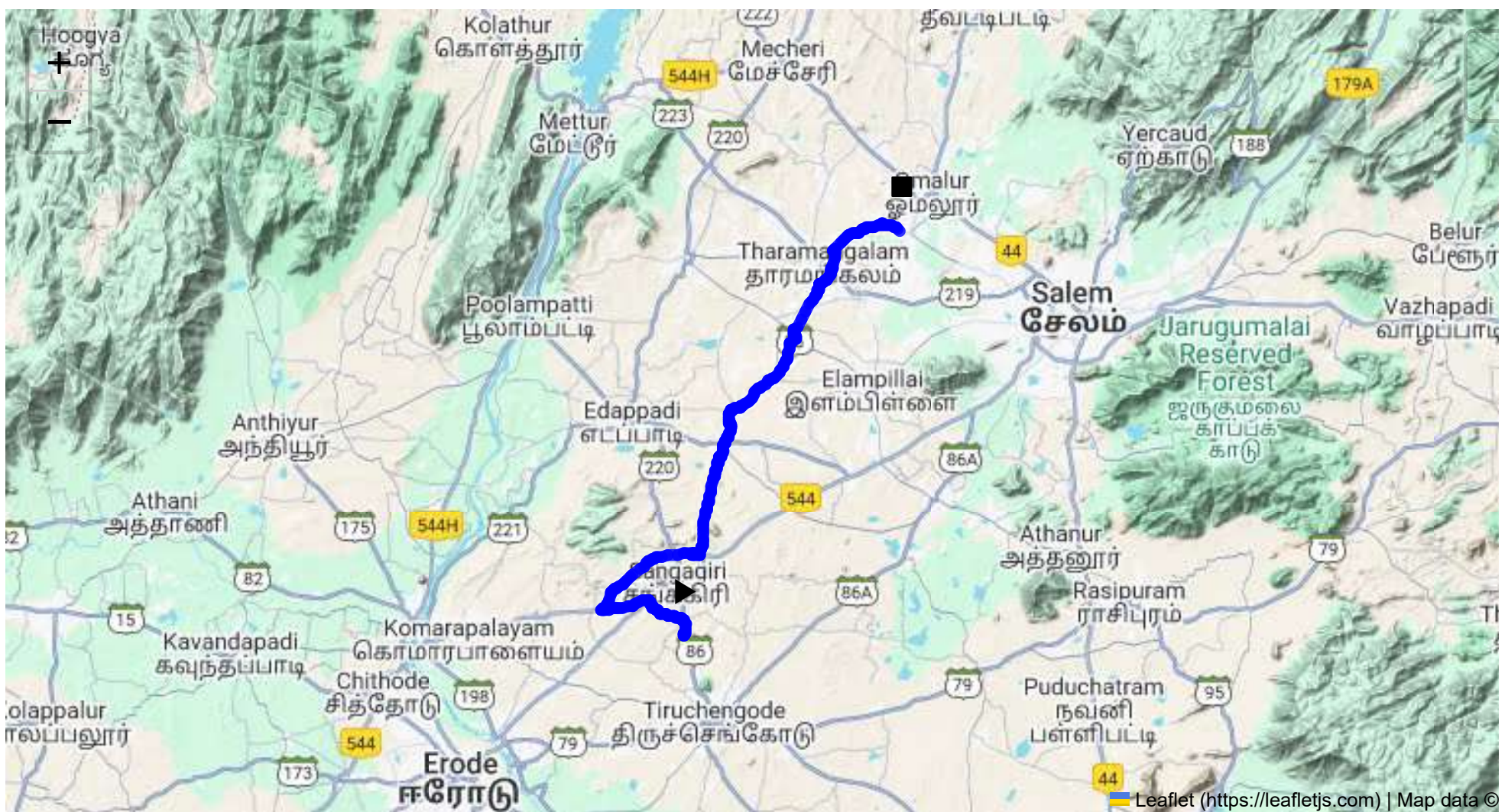
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 50.70 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (11.7158823, 78.02550068)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Salem through the specified waypoints is approximately 50.70 kilometers and traverses several roads, including SH 86. It moves through a mix of urban and semi-urban areas, starting in Sangagiri and concluding in Salem.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region generally experiences a tropical climate:

- **Summer (March to June):** Can be extremely hot, which may affect road grip.
- **Monsoon (July to September):** Roads can become slippery; possible flooding in low areas.
- **Winter (November to February):** Mild; fog may reduce visibility in early mornings.

Potential weather hazards include sharp reductions in visibility during heavy rainfall or fog, especially in early mornings or late evenings.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (8-10 AM) and late afternoon (5-7 PM) see increased traffic, particularly near urban centers.
- **Congestion-Prone Areas:** Major congestion expected in and around Sankari and near the beginnings of Salem. Particularly heavy around town junctions and significant bottlenecks on SH 86.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Generally good on SH 86, though maintenance may vary in rural stretches.
- **Infrastructure:** Adequate signage but limited street lighting in rural stretches. Some areas may lack proper lane markings.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route:** Consider using NH 44, though further analysis of specific segments would be required for emergency detours based on real-time traffic updates and conditions.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Time Restrictions:** Certain sections may have restrictions during peak hours or at night.
- **Permits:** Ensure all necessary permits for transporting hazardous materials are obtained.

7. Overview of Historical Incidents

There have been occasional incidents involving spillage or accidents on SH 86. Most incidents are due to adverse weather or human error.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Proximity to water bodies requires caution to avoid contamination.
- **Environmental Concerns:** Minimize idling in forested segments to reduce air pollution.

9. Analysis of Communication Coverage

- Potential Dead Zones:** Some rural stretches between small villages like Ayveli and Erumaipatti may have gaps in mobile network coverage.

10. Estimated Emergency Response Times

- Urban Areas (e.g., near Salem):** Typically 20-30 minutes.
- Rural Stretches:** Could extend to 45-60 minutes due to limited quick access.

11. Overall Summary of Risk Assessment

- Primary Risks:** Adverse weather, peak-time congestion, narrow roads in some rural areas.
- Mitigation Measures:** Flexible scheduling avoiding peak times and adverse weather; regular communication checks for mobile signals; pre-planned alternative routes.

In summary, while the route is generally navigable, factors such as weather, traffic congestion, and rural road conditions present moderate risks. Proper planning and adherence to local regulations can mitigate many of these risks. Regular updates on traffic and weather conditions are advisable for real-time decision-making.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.45357, 77.85789	30 KM/Hr
4	Turn	Medium	11.45352, 77.85698	30 KM/Hr
5	Turn	Medium	11.46048, 77.85036	30 KM/Hr
6	Turn	Medium	11.46303, 77.84945	30 KM/Hr
7	Turn	Medium	11.46318, 77.84913	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Turn	High	11.45634, 77.81673	15 KM/Hr
10	Turn	High	11.49409, 77.88004	15 KM/Hr
11	Turn	Medium	11.49420, 77.88016	30 KM/Hr
12	Turn	High	11.49383, 77.88526	15 KM/Hr
13	Turn	High	11.49395, 77.88537	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	High	11.49409, 77.88536	15 KM/Hr
15	Turn	Medium	11.49435, 77.88499	30 KM/Hr
16	Turn	Medium	11.49465, 77.88488	30 KM/Hr
17	Turn	Medium	11.56072, 77.89919	30 KM/Hr
18	Turn	Medium	11.56096, 77.89931	30 KM/Hr
19	Turn	High	11.57035, 77.90223	15 KM/Hr
20	Turn	Medium	11.57041, 77.90221	30 KM/Hr
21	Turn	Medium	11.58279, 77.90658	30 KM/Hr
22	Turn	Medium	11.64197, 77.95197	30 KM/Hr
23	Turn	High	11.67868, 77.96836	15 KM/Hr
24	Turn	Medium	11.68904, 77.97736	30 KM/Hr
25	Turn	High	11.68915, 77.97734	15 KM/Hr
26	Turn	Medium	11.70218, 77.97866	30 KM/Hr
27	Turn	High	11.70227, 77.97864	15 KM/Hr
28	Turn	Medium	11.71921, 78.01133	30 KM/Hr
29	Turn	High	11.72035, 78.01270	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46048, 77.85036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45634, 77.81673



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49409, 77.88004



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49420, 77.88016



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88526



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49409, 77.88536



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49435, 77.88499



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56072, 77.89919



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56096, 77.89931



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57035, 77.90223



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57041, 77.90221



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58279, 77.90658



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64197, 77.95197



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67868, 77.96836



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68904, 77.97736



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68915, 77.97734



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70218, 77.97866



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70227, 77.97864



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71921, 78.01133



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.72035, 78.01270
